Provisional Patent Application

Self-Righting Automatic Dumpster Cleaning Device

Background

Dumpsters are a breeding ground for grime, bacteria, and unpleasant odors, creating unsanitary and unpleasant conditions.
Cleaning dumpsters is labor-intensive, time-consuming, and often neglected due to the inconvenience of traditional methods,
which rely on external water sources, manual scrubbing, or costly professional services.

This invention addresses these issues by providing an automated, disposable, and eco-friendly solution that simplifies dumpster maintenance.

Summary of the Invention

The Self-Righting Automatic Dumpster Cleaning Device is a disposable, self-righting device that automates the cleaning process for dumpsters.
It features a weighted base for stability, a pressurized canister containing a biodegradable cleaning solution, and a 360° spray nozzle.
The device activates upon impact or lid closure, spraying foam that adheres to and cleans the walls, floor, and lid of the dumpster.

Detailed Description of the Invention

1. **Outer Shell**:
   - Made of biodegradable plastic or treated cardboard to minimize environmental impact.
   - Dimensions:
     - Residential Model: 6 inches tall, 4-inch diameter.
     - Commercial Model: 10 inches tall, 6-inch diameter.

2. **Weighted Base**:
   - Contains encased recycled metal pellets or sand for self-righting stability.
   - Includes an absorbent layer (e.g., silica gel or activated carbon) to soak up residual liquids in the dumpster.

3. **Cleaning Solution Canister**:
   - Pressurized aluminum or plastic container.
   - Contains a biodegradable cleaning solution designed for foam expansion and

adherence.
- Cleaning Solution Ingredients:
  - Sodium Lauryl Sulfate (40%): Surfactant for grime removal.
  - D-Limonene (20%): Citrus-based degreaser.
  - Hydrogen Peroxide (10%): Antibacterial agent.
  - Dimethyl Ether (30%): Eco-friendly propellant.

4. **360° Spray Nozzle**:
   - Multi-hole nozzle design ensures complete interior coverage of the dumpster.
   - Spray radius:
     - Residential Model: 5 feet.
     - Commercial Model: 8 feet.

5. **Activation Mechanism**:
   - Impact-Triggered: Pressure-sensitive diaphragm bursts upon hitting the dumpster floor.
   - Lid Closure Trigger: Sensor detects lid motion or darkness and activates spray.
   - Timer Backup: Activates spray within 5 seconds if primary triggers fail.

How It Works:
- After trash pickup, the user tosses the device into the empty dumpster.
- The weighted base ensures the device lands upright.
- Spray mechanism activates and distributes foam, cleaning the dumpster's interior.

Claims

1. A disposable cleaning device that self-rights using a weighted base and sprays a biodegradable cleaning solution in a 360° arc.
2. Dual activation mechanisms, including impact triggers and lid closure sensors, for reliable operation in diverse scenarios.
3. Weighted base with integrated absorbent materials for liquid collection and odor reduction.
4. Use of biodegradable and recyclable materials for environmentally friendly disposal.
5. A cleaning solution that adheres to surfaces, neutralizes odors, and sanitizes effectively.

Drawings/Illustrations

N/A

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